



**DUNMAN HIGH SCHOOL  
PRELIMINARY EXAMINATION 2018  
YEAR SIX  
H1 BIOLOGY (8876)**

**Suggested Answers**

**Question 1**

**(a)**

Resolution refers to the degree of details that can be seen whilst magnification refers to the number of times a specimen can be enlarged

**(b)(i)**

I : nuclear envelope

II : mitochondrion

**(ii)**

I :

Contain nuclear pores to allow for RNA to leave nucleus and nucleotides and enzymes to enter nucleus

**(iii)**

II : Involved in cellular (aerobic) respiration that result in the formation of ATP

**(iv)**

Eukaryotic cell due to presence of membrane bound organelles namely mitochondria and nucleus

**(c)**

mRNA

sequence of codons determine the sequence of amino acids in polypeptide  
direct the initiation and termination of translation by the start and stop codon

rRNA

rRNA is a component of ribosome

rRNA component interacts with mRNA and tRNA by bringing them to correct orientation during proteins synthesis

**Question 2**

**(a)(i)**

Only seen in males / not in females

**(a)(ii)**

Unaffected parents/mother → child with M.D. / (1 x) 2 → 5 / (3 x) 4 → 11 / 8 (x 9) → 13

**(b)**

5  $X^{dY}$

6  $X^{DY}$

7  $X^DX^d$  AND  $X^DX^D$

8  $X^DX^d$

All 4 correct = 2 marks

2 or 3 correct = 1 mark

(c)

 $\frac{1}{4}$  / 0.25 / 25% / 1:3 / 1 in 4; (NOT '1:4')**Question 3**

(a)

- A codon is three nucleotides/bases on mRNA that specify an amino acid
- It is where the corresponding tRNA anticodon will complementary base-pairs
- signals the start of translation
- signals the termination of translation

(b)

- Single base pair substitution mutation from guanine to adenine on DNA template strand
- Results in the substitution/replacement of an amino acid from alanine to valine

(c)

- Ref. alanine and valine gives rise to different extent of hydrophobic interaction
- more extensive hydrophobic interaction between  $\beta$  peptides and thus forming amyloid plaques

**Question 4**

(a)

- Eggs are laid on damp surfaces/ surface of stagnant water
- Larvae hatch from eggs upon being submerged in water
- larvae feed on microorganisms and particulate organic matter, shedding their skins three times to be able to grow from first to fourth **instars**
- When the larva has acquired enough energy and size and is in the fourth instar, metamorphosis is triggered, changing the larva into a pupa
- Pupae do not feed, they just change in form until the body of the adult, flying mosquito is formed and emerged

(b)

- *Aedes aegypti* is a mosquito vector
- In the last 50 year, as climate change to warmer and wetter in more countries, mosquito also thrives in these new places
- therefore where the vector thrives so does the dengue virus

(c)

- ref. human activity provide viable habitats e.g. stagnant water for Aedes to thrive
- In colder latitudes or altitudes, urbanisation / large cities provide a warmer habitat

**1 Max****Question 5**

(a)

the calls made by cranes are species specific

the greater the similarity in calls, the closer the relationship between different species is likely to be

(b)

- Neanderthals, modern human and chimpanzee are the most closely related
- because the first 20 amino acid sequences for all three organisms are identical
- subsequent relatedness to Neanderthal is in the order of orang-utan, gorilla and monkey
- because orang-utan has 1 amino acid difference from Neanderthals, gorilla 2 differences and monkey 3 differences
- at least one difference described in detail e.g. orang-utan has 1 amino acid difference from Neanderthals at amino acid 19, a change from R to K in orang-utan

4 max

### Question 6

(a)(i)

$$(8.8\% - 6.9\%) / 20 = 0.10\% \text{ per minute}$$

(a)(ii)

1 mark for quoting appropriate pair of values with units

1 mark for describing appropriate difference (using comparative terms) between solutions A and B

E.g.

- From 0-20mins, percentage O<sub>2</sub> in solution **B** increased from **6.9% to 8.95%** but percentage O<sub>2</sub> in solution A increased from **6.9% to 8.8%**.
- From 0-10min, percentage O<sub>2</sub> in solution **B increased more steeply** from **6.9% to 8.7%** but percentage O<sub>2</sub> in solution A increased from **6.9% to 8.1%**.

OR

- From 0-10min, percentage of O<sub>2</sub> in solution **B has a higher increase of 1.8%** but O<sub>2</sub> increase in solution **A increased by only 1.2%**.
- From 10-20min, percentage O<sub>2</sub> in solution **B increased less steeply** from **8.7% to 8.95%** but percentage O<sub>2</sub> in solution A **increased more steeply** from **8.1% to 8.8%**.

OR

- From 10-20min, percentage of O<sub>2</sub> in solution **B has a lower increase of 0.25%** but O<sub>2</sub> increase in solution **A has a higher increase of 0.7%**.

(a)(iii)

(quote values and describe trend) From 0 – 20 mins, percentage of O<sub>2</sub> in solution C decreased from 6.9% to 6.1%

In the absence of light, no photolysis of water, hence, no oxygen released

As oxygen is used up in respiration

2 max

(b)(i)

**A: ATP / ADP**

**B: reduced NAD / NADH**

(b)(ii)

Anaerobic respiration / lactic acid fermentation;

Pyruvate will be used in anaerobic respiration / alcohol fermentation;

to form ethanol and  $\text{CO}_2$  instead of undergoing link reaction to form acetyl CoA;

**Essay Answers****1(a)**

1. channel proteins with hydrophilic pores found within intrinsic protein molecules
2. Conformation change in carrier protein allows selective passage of molecules across the plasma membrane
3. for facilitated diffusion of ions, large and polar molecules/ for osmosis / water molecules diffuse across the membrane
4. for active transport to pumps to move solutes against concentration gradients
5. receptor proteins for binding of antigens / hormones / external signal molecules
6. to allow an external signal to trigger or initiate reactions within a cell
7. Glycoprotein function in cell-cell adhesion / function in cell junctions / cell-cell recognition

**1(b)**

1. ATP is synthesized by substrate level photophosphorylation, oxidative phosphorylation and photophosphorylation.
2. ATP is synthesized during glycolysis, in the cytoplasm, and during Krebs cycle in the mitochondrial matrix. ;
3. 4 ATP / 2 nett ATP is synthesized per glucose molecule during glycolysis. ;
4. In anaerobic respiration, ATP is synthesized only by substrate level phosphorylation in glycolysis. ;
5. In the Krebs cycle, 2 ATP is synthesized per glucose when succinyl-CoA is converted to succinate. ;
6. NAD and FAD are reduced during glycolysis, link reaction and Krebs cycle. ;
7. Reduced NAD and FAD donates electrons to the electron transport chain on the inner mitochondrial membrane. ;
8. As electrons are transported along a series of electron carriers of progressively lower energy levels, some energy is used to pump  $H^+$  from the matrix to the intermembrane space, against its concentration gradient ;
9. This creates a proton gradient across the inner mitochondrial membrane, driving protons to diffuse down its concentration gradient via ATP synthase on the inner mitochondrial membrane. ;
10. ATP synthase harness the proton motive force for phosphorylation of ADP to ATP, in the mitochondria matrix. ;
11.  $O_2$  is the final electron carrier of the electron transport chain. ;
12. 3 ATP is synthesized per reduced NAD and 2 ATP per reduced FAD. ;
13. ATP is produced during non-cyclic photophosphorylation and cyclic photophosphorylation;
14. During non-cyclic photophosphorylation, photolysis of water donates electrons to the electron transport chain on the thylakoid membrane;
15. Light energy is harvested by photosynthetic pigments and the energy is passed on to the reaction centre chlorophyll;
16. Electron at reaction centre chlorophyll a is excited to a higher energy level;
17. As electrons are transported along a series of electron carriers of progressively lower energy levels, some energy is used to pump  $H^+$  from the stroma to the thylakoid lumen, against its concentration gradient ;
18. This creates a proton gradient across the thylakoid membrane, driving protons to diffuse down its concentration gradient via ATP synthase on the thylakoid membrane. ;

**QWC: at least 2 points each from substrate level photophosphorylation, oxidative phosphorylation and photophosphorylation**

**2(a)****2(b)**

1. Cancer arises from uncontrolled cell division / loss of normal control of cell division
2. Due to mutations which alter the genes that control cell division
3. This causes the rate of cell division to exceed the rate of cell differentiation and cell loss
4. And leads to tumour formation / tumourigenesis
5. Which invades / infiltrates into surrounding tissues and disrupts its normal organisation
6. Cancer cells can metastasise / spread *via* blood or lymph vessels to other organs
7. Exposure to chemical carcinogens that are alkylating agents and electrophiles
8. Such as mustard gas, asbestos, polycyclic hydrocarbons, various pesticides and herbicides, carbon monoxide in cigarette smoke / OVA
9. High energy beams / wavelengths such as ultraviolet radiation / X-rays / gamma rays
10. Some viruses can also cause cancer by causing genetic changes in the cells that will make them cancerous
11. Some people also possess genetic predisposition to cancers, where genes involved in regulating cell division such as proto-oncogenes or tumour suppressor genes become defective
12. A compromised immune system could also lead to abnormal proteins on cancer cells to not be targeted and removed
13. As a person ages, cells are more likely to have accumulated enough changes to the genes / mutations to turn into cancerous cells

**9 Max + QWC**

**QWC: Includes 2 points each for characteristics of cancer and factors that increases chances of cancerous growth**